



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

## SAFETY DATA SHEET

### Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

### 1. CHEMICAL IDENTIFICATION

#### Product Identification

Trade name: PU Gibb

Product name: gibberellic acid 4% W/V EC

Product use: Plant growth regulator

#### Relevant identified uses of the substance/mixture and uses advised against:

Gibberellic acid is a hormone found in plants and fungi. It can be used in agriculture to produce bigger leaves and longer stems, enhances photosynthesis, stimulates seed germination and triggers transitions from the vegetative to the flowering stage.

### 2. HAZARD IDENTIFICATION

Flammable liquid

#### Acute Health Effect

Swallowed:

May cause irritation of the digestive tract. Low hazard for usual industrial handling.

Eye:

Irritating to eyes.

Skin:

May cause skin irritation.

Inhaled:

Vapors may cause drowsiness & dizziness. Low hazard for usual industrial handling.

### 3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: (3*S*,3*aS*,4*S*,4*aS*,7*S*,9*aR*,9*bR*,12*S*)-7,12-dihydroxy-3-methyl-6-methylene-2-oxoperhydro-4*a*,7-methano-9*b*,3-propenoazuleno[1,2-*b*]furan-4-carboxylic acid [IUPAC]



## Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

Common name: gibberellic acid

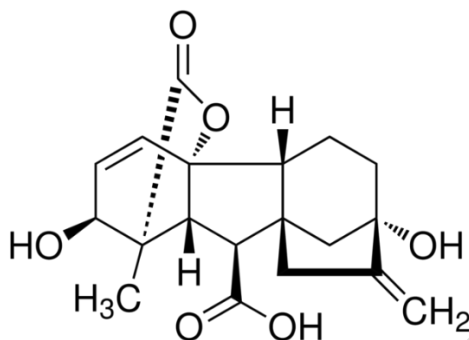
CAS Number: 77-06-5

Ingredients and composition: Active ingredient – gibberellic acid 4% W/V

Formula:  $C_{19}H_{22}O_6$

Molecular weight: 346.379 g/mol

Structural formula:



### 4. FIRST AID MEASURES

#### Description of first aid measures

In case of swallowed:

Do not induce vomiting or give liquids. Seek medical advice.

In case of eyes contact:

Irrigate eyes with running water for at least 15 minutes.

In case of skin contact:

Remove contaminated clothing. Wash the affected area thoroughly with soap and water.

Obtain medical attention if irritation occurs.

In case of Inhaled contact:

Remove the person to a well ventilated area. Ensure the person is warm and comfortable.

Obtain immediate medical attention.

### 5. FIRE FIGHTING MEASURES

Flammability

Flammable

Fire point:

24 °C (PM closed cup)

Explosive properties:

Lower Explosive Limit in air: 2.5%

Upper Explosive Limit in air: 12.7%

#### Extinguishing Media:

Use water mist, fog, CO<sub>2</sub>, foam and/or dry powder.



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

Fire and Explosion hazards:

Flammable liquid and vapor

Products of Combustion and Decomposition:

Carbon compounds.

Fire Fighting Equipment:

Wear full protective clothing and self-contained breathing apparatus.

## 6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Personal Precautions

Wear protective clothing, see Section 8. Exposure control/personal protection. Keep away from sources of ignition.

Environmental precautions

Prevent product or contaminated water entering drains.

Methods for cleaning up

In case of spillage, use inert absorbent materials. Isolate spilled material for safe disposal and collect in sealable polylined containers. Ventilate area. Dispose of contaminated material in accordance with local regulations.

## 7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.

Safe Handling:

Protect against physical damage and store in cool, dry well-ventilated location, away from sources of ignition. Keep away from drink and foodstuffs.

Do not smoke in the workplace. Avoid contact with spray. Wash thoroughly after handling.

Storage Conditions:

Store in dry, cool (below 32°C) conditions in original container. Use flammable liquid storage area.

## 8. EXPOSURE CONTROL/PERSONAL PROTECTION

Exposure standards

To avoid risks to man and the environment comply with instructions for use

Respiratory protection: Avoid inhaling vapor, work in a well ventilated area



## Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

|                  |                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hand protection: | Chemical resistant gloves such as barrier laminate, butyl rubber, nitrile rubber, neoprene rubber, polyvinyl chloride or vinyl should be worn. Refer to product label for more information. |
| Eye protection:  | Glasses/goggles should be worn or face mask used.                                                                                                                                           |
| Skin protection: | Wear sensible regularly laundered overalls. Gloves and boots should be worn.                                                                                                                |

### 9. PHYSICAL AND CHEMICAL PROPERTIES

#### Information on basic physical and chemical properties

|                            |                                       |
|----------------------------|---------------------------------------|
| Physical stage:            | Liquid                                |
| Color:                     | Clear faintly straw                   |
| Odor:                      | Alcoholic                             |
| pH:                        | 3.28 (50% V/V solution)               |
| Viscosity:                 | 3.4503 mm <sup>2</sup> /s             |
| Flash point:               | 24 °C (Pensky-Martens Closed Cup)     |
| Vapor pressure:            | 4.4 kPa at 20 °C                      |
| Vapor density (air =1):    | 2.1                                   |
| Relative density:          | 0.8003 g/ml                           |
| Solubility in water:       | Soluble at all concentrations at 20°C |
| Oxidizing properties:      | Reacts with strong oxidizing agents   |
| Evaporation rate (BuAc=1): | 2.83                                  |
| Other information:         | Auto ignition 445.6°C                 |

### 10. STABILITY AND REACTIVITY

|                                   |                                                                             |
|-----------------------------------|-----------------------------------------------------------------------------|
| Stability:                        | Stable in original containers for at least 3 years at ambient temperatures. |
| Conditions to avoid:              | Excessive heat and direct sunlight. Keep away from sources of ignition.     |
| Materials to avoid:               | Incompatible with strong oxidizing agents.                                  |
| Hazardous decomposition products: | None under normal conditions of storage and use.                            |

### 11. TOXICOLOGICAL INFORMATION

#### Information on Acute toxicity for end product, Gibberellic acid:

|                        |               |
|------------------------|---------------|
| LD50 (Oral, rat):      | > 5,050 mg/kg |
| LD50 (Dermal, rabbit): | > 2,020 mg/kg |



## Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

LC50 (Inhalation, rat, 4h): > 5.23 mg/L

Skin irritation (rabbit): Slight irritant.

Eye irritation (rabbit): Highly eye irritant.

Skin sensitization: Not classified as sensitizer

Respiratory tract: Studies have revealed no observable abnormalities.

Corrosively: Non-corrosive

Chronic toxicity: No studies available

Carcinogenicity data: None (active ingredient)

Mutagenicity data: None (active ingredient)

Reproductive data: No studies available

Note: Gibberellic acid is a plant growth hormone that has been reported to have estrogenic and androgenic activity in animals.

## 12. ECOLOGICAL INFORMATION

### Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the active ingredient, Gibberellic acid:

#### **Fish – Acute toxicity:**

Not toxic but avoid contamination of water courses.

Oncorhynchus mykiss,

LC50 (96 h): >112mg/L

Daphnia magna,

EC50 (48 h): > 122 mg/L Not toxic but avoid contamination of water courses.

#### **Birds – Acute toxicity:**

Colinus virginianus,

LD50: 2,250 mg/kg

#### **Algae – Acute toxicity:**

LC50 (48h): >100 mg/L

NOEC: ≥100 mg/L

#### **Bee – Acute toxicity:**

Apis mellifera,

LC50 (48 h): > 25µg/bee

### Persistence and degradability

Active ingredient is non-toxic and degrades rapidly under biological conditions.



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

### 13. DISPOSAL CONSIDERATIONS

Dispose of empty, used containers

Product:

May be taken to waste disposal site or incineration plant when in compliance with local regulations refer to the local waste regulation authority.

Packaging:

Empty containers can be landfilled when in accordance with local regulation.

### 14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

Toxicity Class: WHO (a.i.) U; EPA III (active ingredient)

UN No: 1993

Dangerous Goods Class: 3.2

Packaging Group: II

Marine pollutant: No

Other applicable information: IATA Class III / ADR Class III

### 15. REGULATORY INFORMATION

Symbol(s): Xi Irritant  
F Highly Flammable

Risk Phrases:

R11: Highly flammable

R36: Irritating to eyes

R67: Vapors may cause drowsiness & dizziness

Safety Phrases:

S2: Keep out of reach of children

S7: Keep container tightly closed

S13: Keep away from food, drink and animal feed stuffs

S16: Keep away from source of ignition – No smoking

S20: When using do not eat or drink

S24/25: Avoid contact with skin & eyes

S26: In case of contact with eyes, rinse immediately with plenty of water and seek medical advice

S36/37/39: Wear suitable protective clothing, gloves and eye/face protection

S51: Use only in well ventilated areas.



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

Specific provisions Restricted to professional users. To avoid risks to man and the environment, comply with the directions for use.

#### **16. OTHER INFORMATION**

For further information on the product and product use refer to the product label.

THEP WATANA CO., LTD.